

Chapter 6, Problem 67P

Problem

An op amp integrator has $R = 50 \text{ k}\Omega$ and $C = 0.04 \text{ }\mu\text{F}$. If the input voltage is $v_i = 10 \sin 50t \text{ mV}$, obtain the output voltage. Assume that at t equal to zero, the output is equal to zero.

Step-by-step solution

Step 1 of 1

Given that,

$$R = 50 \text{ k}\Omega \text{ and}$$

$$C = 0.04 \text{ }\mu\text{F}$$

Therefore,

$$V_o = -\frac{1}{RC} \int V_i dt$$

$$RC = 50 \times 10^3 \times 0.04 \times 10^{-6}$$

$$RC = 2 \times 10^{-3}$$

$$V_o = \frac{-10^3}{2} \int 10 \sin 50t \, dt$$

$$\therefore \boxed{V_o = 100 \cos 50t \text{ mV}}$$